

1. Functional Requirements

ID	Requirement	Acceptance condition
FR-01	Accept document type	User can select a supported type or enter a custom type.
FR-02	Accept party details	Parties can be entered as free-form text.
FR-03	Accept dates	Effective-date/date details are required.
FR-04	Accept terms	Terms and conditions can be entered as semicolon-separated items.
FR-05	Generate draft	POST /generate returns a structured document when valid input is supplied.
FR-06	Preview/edit	Generated text is displayed and can be reviewed or edited.
FR-07	Export	Draft can be exported to TXT, DOCX and PDF.
FR-08	Demo mode	Application can generate a deterministic local demonstration document without live Gemini generation.

2. Non-Functional Requirements

Category	Requirement
Usability	Keep the main workflow within one workspace and show missing required fields clearly.
Security	Sanitize input and escape preview HTML; do not expose backend mechanics in the user interface.
Reliability	Retry temporary provider failures and try additional accessible models.
Maintainability	Separate UI, API routes, AI generation, text utilities and export services.
Portability	Support local execution through Python/virtual environment and standard commands.
Testability	Provide automated tests for API, validation, text utilities and exporters.

3. Input / Processing / Output Model

Layer	Data
Input	document_type, parties, terms, dates
Validation	Pydantic field constraints + sanitize_text()
Generation	GeminiDocumentGenerator.build_prompt() + model selection/fallback
Application response	document, document_type, demo_mode, model
Output	Preview plus TXT, DOCX and PDF download formats

4. API Contract

POST /generate

```
{"document_type":"...", "parties":"...", "terms":"...", "dates":"..."}
```

```
Response: {"document":"...", "document_type":"...", "demo_mode":false, "model":"..."}
```

5. Validation Rules

- The four request fields are mandatory and reject empty / whitespace-only content.

- Each field has configurable maximum length limits.
- Extra JSON fields are forbidden by the Pydantic model configuration.
- Text is normalized to remove problematic control characters and normalize punctuation.
- The browser preview escapes HTML before rendering supported Markdown syntax.

6. Constraints and Assumptions

- A working Gemini API key is required for live AI generation.
- The generated document is a draft and requires human review.
- Legal requirements vary by jurisdiction and transaction; the system does not independently verify them.
- The supplied project package is designed for a local academic demonstration and does not include a production database or authentication layer.